Fall 2013 / EC571

Instructor:Ajay Joshi

Final Exam

Date: Dec 16, 2013

Name : _____

BU ID : _____

Question	Maximum points	Points scored
1	20	
2	20	
3	20	
4	20	
5	20	
6	20	
Bonus	5	
Total	100	

1. The duration of the exam is 2 hours
2. The maximum grade is 100 points
3. Solve any 5 questions. If you solve all 6 questions, you need to indicate which 5 problems should be graded.
4. Please write your name on all pages
5. The solutions should be written in the spaces provided on the test sheet
6. If you need to use extra sheets of paper, write your name and BU ID on it, and clearly indicate the question number
7. Material sharing is not allowed
8. Calculators are allowed
9. Boston University Academic Code of Conduct applies

Question 1

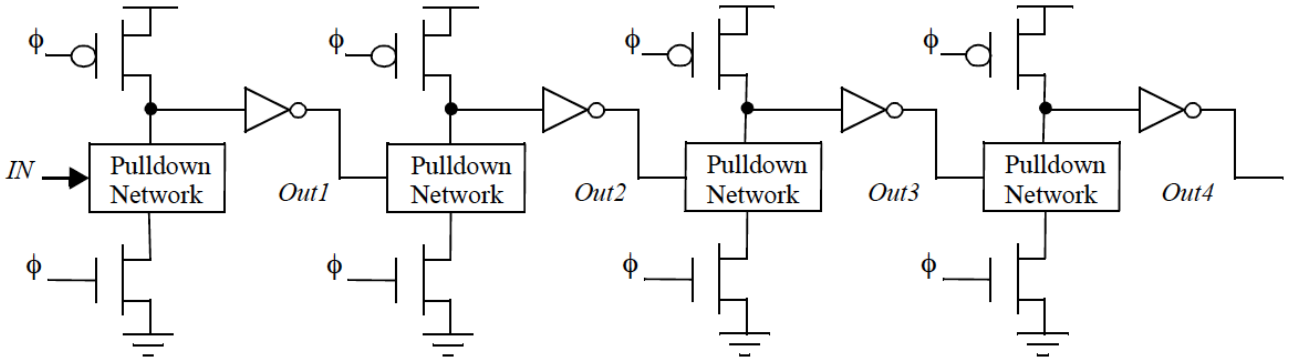


Figure 1

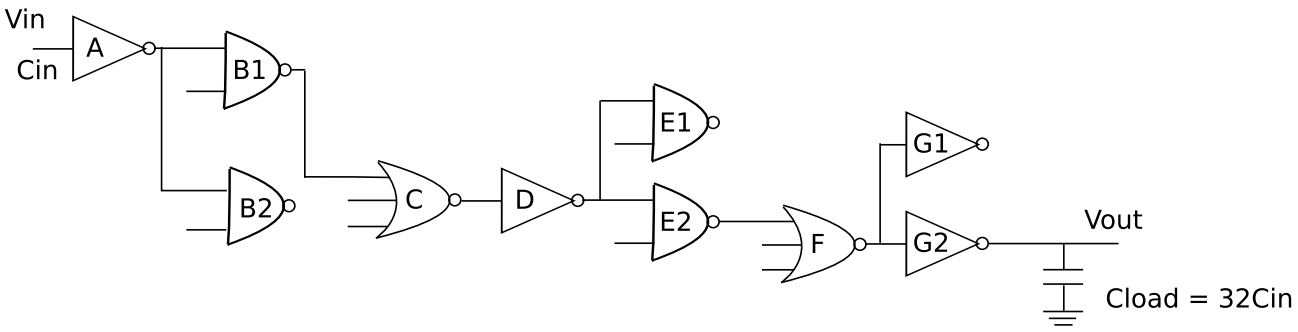
Consider a conventional 4-stage Domino logic circuit as shown in Figure 1 in which all precharge and evaluate devices are clocked using a common clock ϕ. For this entire problem, assume that the pulldown network is simply a single NMOS device, so that each Domino stage consists of a dynamic inverter followed by a static inverter. Assume that the precharge time, evaluate time, and propagation delay of the static inverter are all T/2. Assume that the transitions are ideal (zero rise/fall times).

1. Complete the timing diagram for signals Out_1, Out_3, Out_3 and Out_4 for one clock period, when the *IN* signal goes high before the rising edge of the clock ϕ. Assume that the clock period is 10 T time units. (10 points)

2. Suppose that there are no evaluate switches at the 3 latter stages. Assume that the clock ϕ is initially in the precharge state (ϕ=0 with all nodes settled to the correct precharge states), and the block enters the evaluate period (ϕ=1). Is there a problem during the evaluate period, or is there a benefit? Explain. (5 points)

3. Assume that the clock is initially in the evaluate state (ϕ=1), and the block enters the precharge state (ϕ = 0). Is there a problem, or is there any benefit, if the last three evaluate switches are removed? Explain. (5 points)

Question 2



For the circuit shown in the above figure, using the logical effort technique, calculate the input width of transistors for each gate in the path such that total delay of the path is minimized. You can assume that gates B1 and B2 are identical, gates E1 and E2 are identical and gates G1 and G2 are identical. Assume Cin = 4 fF, Cg = 1 fF/u. (20 points)

Question 3

Design the static complementary CMOS implementation of a 3-bit Gray encoder circuit (see truth table below) for minimum delay (assuming a stage effort of 4) with a load of 20 fF. As part of the design you need to determine the width of all transistors. You can use the following transistor parameters for the design - L = 0.25 micron, W_{min} (minimum transistor width) = 0.3 micron $R_{sq}(NMOS)$ = 13 Kohm/sq, $R_{sq}(PMOS)$ = 26 Kohm/sq, C_g = 1.5 fF/micron, C_d = 2 fF/micron. You can assume that the input signals have very small rise and fall times. Also note that if the transistor values that you calculated are less than the minimum size, then you have to use the minimum size widths. (20 points)

Table 1: Truth table for Gray Encoding

Conventional Binary Encoded Value	Gray Encoded Value
$b_2\ b_1\ b_0$	$g_2\ g_1\ g_0$
0 0 0	0 0 0
0 0 1	0 0 1
0 1 0	0 1 1
0 1 1	0 1 0
1 0 0	1 1 0
1 0 1	1 1 1
1 1 0	1 0 1
1 1 1	1 0 0

Question 4

Sketch the schematics for $f(A,B,C) = \overline{A \cdot B + C}$ using each of the following circuit techniques. You can assume that you have both complemented and non-complemented inputs:

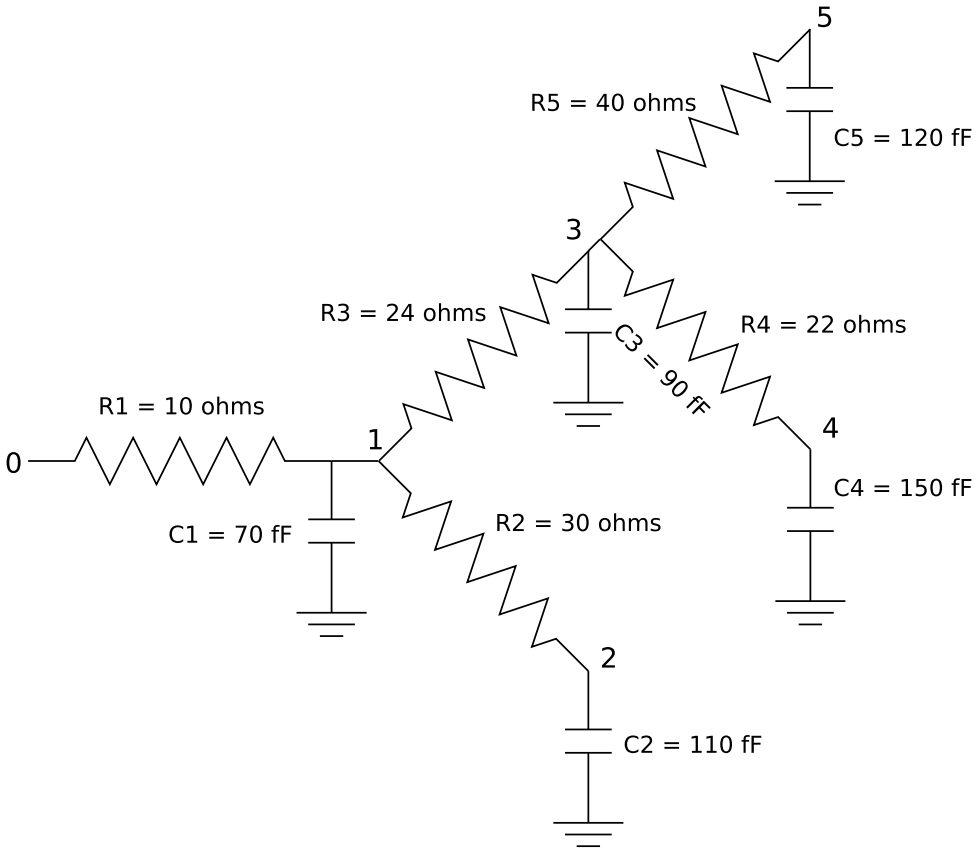
1. Static CMOS. (5 points)

2. Pseudo-nMOS. (5 points)

3. Dual-rail domino. (5 points)

4. Pass transistors. (5 points)

Question 5



1. Write the Elmore delay equations for paths from node 0 to node2, node 0 to node 4 and node 0 to node5. (7 points)

2. Calculate the delay for the paths from node 0 to node2, node 0 to node 4 and node 0 to node5. (7 points)

3. Calculate the energy consumed when the input at node 0 transitions from logic 0 to logic 1 and this logic 1 is transmitted along the three paths. You can assume logic 1 corresponds to a voltage of 1.5 V and logic 0 corresponds to a voltage of 0 V. (6 points)

Question 6

Consider the circuit shown below. Assume a clk-to-q delay of 2 units for the register and a unit delay for each logic block. The registers are positive-edge triggered and have a setup time of 2 units. The multiplexer has 4 units of delay.

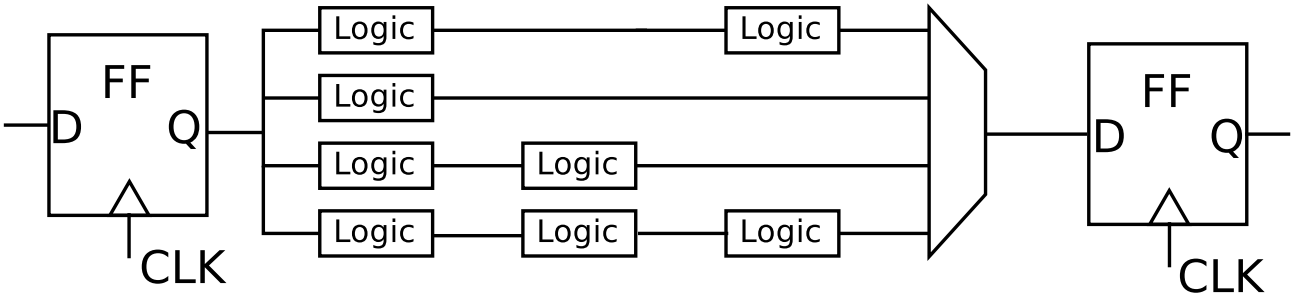


Figure 2

1. Determine the minimum clock period for the above circuit. You can ignore clock skew. (5 points)

2. Determine the maximum hold time for the above circuit. You can ignore clock skew. (5 points)

3. Determine the minimum clock period considering a negative clock skew of 3 time units. (5 points)

4. Determine the maximum hold time considering a negative clock skew of 3 time units. (5 points)

Bonus question

What are the four fundamental two-terminal circuit elements? How do they represent the relationship between voltage, current, flux and charge?